

Limitations of Self-Submitted Admissions Data

Laila Afmege — JHED: lafmege1 — Module 3 Written Reflection

Grad Cafe's dataset is built entirely from applicants who chose, unprompted, to visit the site and type in their own results, which means every statistic computed from it describes the population of people who self-select into reporting — not the population of everyone who applied. That gap shows up directly in our own numbers: PhD entries outnumber Masters entries by roughly three to one in this dataset (21,243 vs. 7,640), a ratio far more lopsided than is plausible for actual national graduate admissions, where Masters programs enroll far more students overall. The likeliest explanation isn't that fewer people pursue Masters degrees; it's that Grad Cafe's userbase, culture, and word-of-mouth reach skew toward research-track PhD applicants who are more invested in tracking admissions cycles closely and comparing notes with peers. Missing values compound this problem rather than randomly diluting it: GPA is present for only about 60% of entries and GRE scores for well under 10%, and there is no way to know whether the applicants who omitted a GPA did so because it was unremarkable, because they were embarrassed by it, or simply because the site's submission form doesn't require it. Any average computed only over the applicants who did report a value — which is what every average in this project necessarily is — may be systematically shifted by whichever kind of person is more likely to fill in that field.

Beyond who chooses to submit at all, self-reported values are also not always reliable once entered, and our own data surfaced a concrete example of this: 1,416 of 2,373 raw GRE Quantitative entries (roughly 60%) fell outside the valid 130-170 ETS scale, with values as high as 999 and clusters in the 300-340 range that look like combined Quantitative-plus-Verbal scores rather than a Quantitative score alone. We chose to exclude those out-of-scale values from the reported average rather than let them distort it, but that decision itself illustrates the limitation: we cannot know, from the data alone, whether a given entry is a genuine typo, a deliberate combined score, or something else, and different plausible cleaning choices would produce different published averages. A similar reliability question applies to outcomes themselves — our analysis found that applicants with a GPA at or above 3.8 were accepted at a very slightly lower rate (39.29%) than applicants below 3.8 (43.74%), a result that is almost certainly confounded by which programs each group applied to (stronger applicants likely target more selective PhD programs) rather than evidence that a higher GPA hurts admission odds. It's also worth asking whether applicants with unusually good or unusually bad outcomes are more

likely to post in the first place, or to post with more detail and emotional investment in the comments field, further skewing any statistic that treats every entry as an equally representative data point. Taken together, this means our database can tell us, with reasonable confidence, what Grad Cafe's self-selected posters reported and when — but it cannot, on its own, support claims about the true admissions rates, average qualifications, or experiences of the broader applicant population that never appears in this dataset at all.